

Exercise Set 4 - Part 2

Problem 4

Pfeifer/Stephan (2019) investigate whether there are gender differences in the fairness perception of one's own wage. This is particularly relevant with respect to negotiations about subsequent wage growth. Table 2 shows average marginal effects for the probability to perceive the own wage as fair resulting from random effects binary probit regressions.

Table 2: Average marginal effects from probits for the probability to perceive the own wage as fair

	(1)	(2)
Female	0.0454 (0.0126)	
Female $\times$ single		0.0263 (0.0200)
Female $\times$ married		0.0498 (0.0156)
Hourly net wage in Euros	0.0353 (0.0015)	YES
Actual weekly working hours	-0.0062 (0.0005)	YES
Marital status		YES
Number of observations	26,051	23,216
Number of individuals	12,213	10,967

Note: Standard errors are in parentheses. Sample includes only singles and married (not separated) persons.

Source: Pfeifer, C., Stephan, G. (2019). Why women do not ask: gender differences in fairness perceptions of own wages and subsequent wage growth. Cambridge Journal of Economics, 43:295–310.

4.1 Name a reason why the authors prefer a random effects model over a fixed effects model?

- The authors are interested in the time-invariant gender (biological sex) variable. This variable has no within variance to exploit in the fixed effects model.
- Conditional fixed effects model is difficult to interpret and based on a restricted data set.

4.2 Interpret the estimation results for the three main explanatory variables in specification (1).

- Female: women are on average 4.5 percentage points more likely to perceive their wages as fair than men.
- Hourly wage: increases probability to perceive own wage as fair by 3.5 PP per Euro.
- Weekly working hours: One working hour more per week decreases the prob of perceiving own wage as fair by 0.6 pp.

4.3 Specification (2) reports interaction terms of female with being single or married. Is there a difference between married and single women?

- Married women are on average 5.0 percentage points more likely to perceive their wages as fair than single men.
- Single women only have a 2.6 percentage points higher probability than single men ($p=0.19$).
- The gender gaps are larger for married women than for single women.

Problem 5

Lemieux et al. (2009) analyse wages in performance-pay jobs. Using panel data for workers in the US, they estimate OLS and fixed effects regressions of log wages on a dummy for whether or not the worker receives performance-pay (e.g. bonuses, stock options, commissions). The model also includes experience (number of years in the workforce, measured in 20 years) and tenure (number of years at this specific job, measured in 10 years) as well as further control variables. Table 3 shows the estimation results for the coefficients. Standard errors are in parentheses.

Table 3: Skills-related wage differentials and performance-pay (PP) Jobs

	(1) OLS	(2) FE
Performance-pay job dummy	-0.4526 (0.1019)	-0.2061 (0.0723)
Education (measured in years)	0.0637 (0.0040)	0.0167 (0.0091)
Education $\times$ performance-pay job	0.0365 (0.0071)	0.0169 (0.0048)
Experience (measured in 20 years)	0.3010 (0.0294)	0.4545 (0.1258)
Experience $\times$ performance-pay job	0.1162 (0.0584)	0.0149 (0.0501)
Tenure (measured in 10 years)	0.2262 (0.0154)	0.1158 (0.0129)
Tenure $\times$ performance-pay jobs	-0.0666 (0.0301)	0.0278 (0.0237)

Note: Column (1) shows results from an OLS regression. Column (2) shows results from a fixed effects regression (worker fixed effect). Dependent variable is log wage. Standard errors are in parentheses.

Source: Lemieux, T., Macleod, W. B., and Parent, D. (2009). Performance pay and wage inequality. *The Quarterly Journal of Economics*, 124(1):1–49.

5.1 Consider the results in column (1). Are the returns to education higher for performance-pay jobs than for non-performance-pay jobs? Provide and interpret the difference in returns. Assess whether the difference is statistically significant.

- Difference in returns due to performance-pay jobs is 0.0365 (log points).
- Interpretation: the returns to education are about 3.7 percentage points higher for workers in performance-pay jobs than that for workers in non-performance-pay jobs.
- The difference is statistically significant at the 5% level since the t statistic for the test that the coefficient is equal to zero is $0.0365/0.0071 = 5.214 > 2$.

5.2 Consider the results in Table 3, column (1). What is the effect of performance-pay for a worker with 16 years of education, 20 years of experience, and 10 years of tenure? Show your calculations and give an interpretation.

- Calculation: the difference in log wages between a performance-pay job and a non-performance-pay job for a person with the given characteristics is given by

$$\beta_{pp} + 16 \cdot \beta_{ed,pp} + 1 \cdot \beta_{exp,pp} + 1 \cdot \beta_{ten,pp} = -0.4526 + 16 \cdot 0.0365 + 1 \cdot 0.1162 + 1 \cdot (-0.0666) = 0.181$$

- The worker will earn approximately 18% (exactly $[\exp(0.181) - 1] \cdot 100\% = 19.7\%$) higher wages in a performance-pay job than in a non-performance job.

5.3 The estimate of years of education falls from 0.0637 in column (1) to 0.0167 in column (2). Provide an explanation for this finding. Assume that ability is an unobserved variable.

- Generally, omitted variable bias arises if a variable that is correlated with the included variables is excluded from the model.
- Unobserved ability is an omitted variable. Ability is expected to be (positively) correlated with education.
- Therefore, deletion of ability from the OLS regression will lead to biased estimates of the parameter for education.
- The OVB problem can be solved by introducing a worker-specific effect in the fixed effects regression, assuming that ability is time-invariant.

References

- Lemieux, T., Macleod, W. B., and Parent, D. (2009). *Performance pay and wage inequality*. The Quarterly Journal of Economics, 124 (1), 1-49.
- Pfeifer, C., Stephan, G. (2019). *Why women do not ask: gender differences in fairness perceptions of own wages and subsequent wage growth*. Cambridge Journal of Economics, 43:295–310.
- Riphahn, R., Wambach, A. and Million, A. (2003). *Incentive effects in the demand for health care: a bivariate panel count data estimation*. Journal of applied econometrics, 18:397-405.